

GED Math Unit: Decimals and Fractions

GED Preparation

November 6, 2025

Unit Overview

- 1 Lesson 1: Understanding and Comparing Decimals
- 2 Lesson 2: Operations with Decimals
- 3 Lesson 3: Understanding Fractions
- 4 Lesson 4: Operations with Fractions
- 5 Lesson 5: Converting Between Fractions and Decimals
- 6 Lesson 6: Applications of Decimals and Fractions
- 7 Exit Ticket & Practice

Lesson 1 Objectives

- Understand the place value of decimals (tenths, hundredths, thousandths).
- Compare and order decimals.
- Convert between decimals and fractions.

Lesson 1 — Key Ideas

- Place value chart: 1.234 has 3 in the hundredths place and 4 in the thousandths place.
- Compare decimals using place value and by aligning digits with zeros as placeholders.
- Convert decimals to fractions by reading the place value (e.g., $0.45 = \frac{45}{100}$).

Lesson 1 — Worked Examples

- 1 Write 0.45 in words: **forty-five hundredths**.
- 2 Which is greater: 0.7 or 0.65? Compare tenths: 7 tenths vs 6 tenths $\Rightarrow 0.7 > 0.65$.
- 3 Convert 0.25 to a fraction: $0.25 = \frac{25}{100} = \frac{1}{4}$.
- 4 Convert $\frac{3}{10}$ to a decimal: 0.3.
- 5 Order 0.56, 0.5, 0.506 by rewriting as 0.560, 0.500, 0.506 $\Rightarrow 0.5, 0.506, 0.56$.

Lesson 2 Objectives

- Add and subtract decimals by aligning decimal points.
- Multiply and divide decimals by tracking decimal places.

Lesson 2 — Strategy Notes

- **Addition/Subtraction:** line up the decimal points; use zeros as placeholders.
- **Multiplication:** multiply as whole numbers, then place the decimal; the total decimal places equals the sum from the factors.
- **Division:** move the decimal in the divisor to make it a whole number and move the dividend's decimal the same amount.

Lesson 2 — Worked Examples

- ① $4.52 + 3.6 = 4.52 + 3.60 = \mathbf{8.12}.$
- ② $5.7 - 2.45 = 5.70 - 2.45 = \mathbf{3.25}.$
- ③ $0.4 \times 0.6 = 4 \times 6 = 24$ with two decimal places $\Rightarrow 0.24.$
- ④ $6.3 \div 0.9 \Rightarrow 63 \div 9 = \mathbf{7}.$
- ⑤ $0.25 \times 12 = 25 \times 12 = 300$ with two decimal places $\Rightarrow 3.00.$

Lesson 3 Objectives

- Identify numerator and denominator.
- Simplify fractions using greatest common factor (GCF).
- Find equivalent fractions.

Lesson 3 — Key Ideas

- A fraction $\frac{a}{b}$ represents a parts out of b equal parts ($b \neq 0$).
- To simplify, divide numerator and denominator by their GCF.
- Create an equivalent fraction by multiplying (or dividing) top and bottom by the same nonzero number.

Lesson 3 — Worked Examples

- 1 Parts of $\frac{3}{5}$: numerator = 3, denominator = 5.
- 2 Simplify $\frac{8}{12}$: $\text{GCF} = 4 \Rightarrow \frac{8}{12} = \frac{2}{3}$.
- 3 Equivalent to $\frac{1}{2}$ with denominator 10: $\frac{1}{2} = \frac{5}{10}$.
- 4 Simplify $\frac{45}{60}$: $\text{GCF} = 15 \Rightarrow \frac{45}{60} = \frac{3}{4}$.
- 5 Are $\frac{4}{8}$ and $\frac{1}{2}$ equivalent? Yes, $\frac{4}{8} \rightarrow \frac{1}{2}$.

Lesson 4 Objectives

- Add and subtract with common denominators (or find the LCD).
- Multiply numerators and denominators; simplify.
- Divide by multiplying by the reciprocal.

Lesson 4 — Strategy Notes

- **Addition/Subtraction:** Use the least common denominator (LCD); convert, then add/subtract numerators.
- **Multiplication:** Multiply across; reduce before or after.
- **Division:** Keep-Change-Flip: keep the first fraction, change to multiply, flip the second.

Lesson 4 — Worked Examples

① $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}.$

② $\frac{3}{8} - \frac{1}{8} = \frac{2}{8} = \frac{1}{4}.$

③ $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12} = \frac{1}{2}.$

④ $\frac{5}{6} \div \frac{2}{3} = \frac{5}{6} \times \frac{3}{2} = \frac{15}{12} = \mathbf{1\frac{1}{4}}.$

⑤ $\frac{1}{2} + \frac{1}{3}$: LCD=6 $\Rightarrow \frac{3}{6} + \frac{2}{6} = \frac{5}{6}.$

Lesson 5 Objectives

- Convert fractions to decimals via long division.
- Convert terminating and repeating decimals to fractions.

Lesson 5 — Key Ideas

- Fractions with denominators made of 2s and 5s (like 8, 20, 125) produce terminating decimals.
- Repeating decimals represent rational numbers: $0.\overline{3} = \frac{1}{3}$.
- To write a decimal as a fraction, read its place value and simplify.

Lesson 5 — Worked Examples

① $\frac{1}{4} \Rightarrow 1 \div 4 = 0.25.$

② $\frac{2}{5} \Rightarrow 2 \div 5 = 0.4.$

③ $0.75 = \frac{75}{100} = \frac{3}{4}.$

④ $0.\overline{3} = \frac{1}{3}$ (standard result for repeating 3).

⑤ $0.875 = \frac{875}{1000} = \frac{7}{8}.$

Lesson 6 Objectives

- Solve real-world word problems using decimals and fractions.
- Interpret and compute with measurements, money, and ratios.

Lesson 6 — Worked Examples

- 1 A shirt costs \$19.99 and tax is \$1.60. Total cost: $\$19.99 + \$1.60 = \mathbf{\$21.59}$.
- 2 You ran $\frac{3}{4}$ mile and your friend ran $\frac{1}{2}$ mile. Difference: $\frac{3}{4} - \frac{1}{2} = \frac{1}{4}$ mile.
- 3 Find 0.6 of 50: $0.6 \times 50 = \mathbf{30}$.
- 4 Recipe uses $\frac{1}{2}$ cup sugar and $\frac{1}{4}$ cup flour. Total: $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ cup.
- 5 You have \$5 and spend \$3.25. Remaining: $\$5.00 - \$3.25 = \mathbf{\$1.75}$.

Quick Check (Exit Ticket)

- 1 Write 0.306 as a fraction in simplest form.
- 2 Order: 0.68, 0.6, 0.605.
- 3 Compute: $3.75 - 1.8$.
- 4 Simplify: $\frac{18}{24}$.
- 5 Compute: $\frac{2}{5} + \frac{3}{10}$.

Guided Practice (Optional)

① $0.07 + 0.6$

② 1.2×0.3

③ $\frac{4}{9} \times \frac{3}{8}$

④ $\frac{7}{10} \div \frac{1}{5}$

⑤ Convert 0.2 to a fraction

⑥ Convert $\frac{9}{20}$ to decimal

Summary

- Decimals and fractions represent parts of a whole and convert between each other.
- Operations follow consistent rules: align decimals; use LCDs; multiply across; divide by reciprocal.
- Apply these skills to money, measurement, and data contexts on the GED.